



International Conference on Data Science, AI and Applications (ICDSAIA 2025)



SPRINGER NATURE

Paper ID – 151, Technical Session – 23

BanglaSentNet: A Hybrid Deep Learning Framework for Multi-Aspect Sentiment Analysis in Bangla E-Commerce Reviews

Ariful Islam and Md Rifat Hossen

Chittagong University of Engineering and Technology

EATL Innovation Hub

18 July 2025

Overview

- Introduction & Problem Statement
- Related Work & Background
- Dataset & Preprocessing
- BanglaSentNet Architecture
- Training & Ensemble Strategy
- Results & Performance Analysis
- Conclusion & Future Work
- References

Introduction

- Bangla (Bengali) is spoken by over 300 million people, yet it has **scarce NLP resources**.
- E-commerce reviews in Bangla often contain **multiple opinions** on various aspects (quality, price, service).
- Traditional sentiment analysis models miss these **nuanced aspects**, predicting only single polarity.
- Challenges include **limited benchmarked datasets** and preprocessing tools for Bangla text [6].



Problem & Challenges

- E-commerce reviews often contain **mixed sentiments** towards different aspects.
- Goal: Perform **multi-aspect sentiment classification** in Bangla across categories: Quality, Service, Pricing, Decoration.
- Challenges: Complex Bangla morphology, informal writing, spelling variations, and code-mixing [6].
- Limited pre-trained models are fine-tuned specifically for Bangla [12].

Mixed Sentiment Example

Original: “পণ্য মান খুব ভালো, কিন্তু দাম অনেক বেশি।”

Translation: “Product quality is very good, but price is too high.”

Analysis: Quality (**Positive**), Price (**Negative**)

Key Contributions



Dataset Creation

Large Bangla e-commerce review dataset labeled by aspect (Quality, Service, Price, Decoration).



Ensemble Strategy

Weighted ensemble strategy combining models to improve classification accuracy.



Hybrid Deep Model

Design integrating LSTM, BiLSTM, GRU, and BanglaBERT (transformer) for enhanced learning.



Data Preprocessing

Feature engineering to handle Bangla-specific issues: class imbalance, misspellings, informal text.

Related Work – International & ABSA Foundations

International ABSA Foundations

- SemEval tasks established ABSA benchmarks [1,2].
- Advanced techniques include **Hierarchical GCNs** for explicit and implicit sentiment [3].
- Hybrid **lexical-ML approaches** are used in service industry ABSA [4].

Early Bangla NLP

- Mandal et al. pioneered supervised learning for **Bangla** classification [5].
- Dhar et al. identified challenges like **morphological complexity** in ML classifiers [6].
- Neural transition: Multi-label classification using **Bengali word embeddings** [7].

Related Work – Deep Learning & Modern Approaches

Deep Learning Advances

- CNN, BiLSTM, GRU models compared for Bangla SA [8].
- Transfer learning with Multilingual BERT shown effective for Bengali SA [9].
- Specialized datasets like **BanglaBook** created for domain-specific research [10].

Transformer & Ensemble Methods

- BERT and BanglaBERT applied to low-resource Bengali tasks [11,12].
- Hybrid approaches combining Lexicon + BanglaBERT achieved record performance [13].

Dataset and Annotation

- **Data Collection:** We collected 8,755 Bangla product reviews from prominent e-commerce platforms, including Daraz, Facebook Marketplace, Rokomari, and Shajgoj.
- **Manual Annotation:** Trained Computer Science students manually labeled each review, with domain experts ensuring high-quality annotations for Quality, Service, Pricing, and Decoration (Positive, Negative, or Neutral).

Table 1: Reviews by Platform

Platform	Review Count
Daraz	6,521
Facebook	1,145
Rokomari	494
Shajgoj	155
Other	440
Total	8,755

Table 2: Sentiment Distribution by Aspect

Aspect	Positive	Negative	Neutral
Quality	2,400	1,200	400
Service	600	300	100
Price	1500	900	100
Decoration	750	400	105
Total		8,755	

Preprocessing Pipeline



Automated Cleaning

Normalized text by removing noise, special characters, and non-Bangla scripts.
Corrected common spelling variants using custom Bangla tokenizers.



Result

A linguistically consistent and high-quality dataset comprising 8,755 product reviews, ready for model training and evaluation.



Manual Review

Eliminated spam, emojis, and poorly structured reviews. Verified terminology using a comprehensive Bangla dictionary to ensure data integrity.

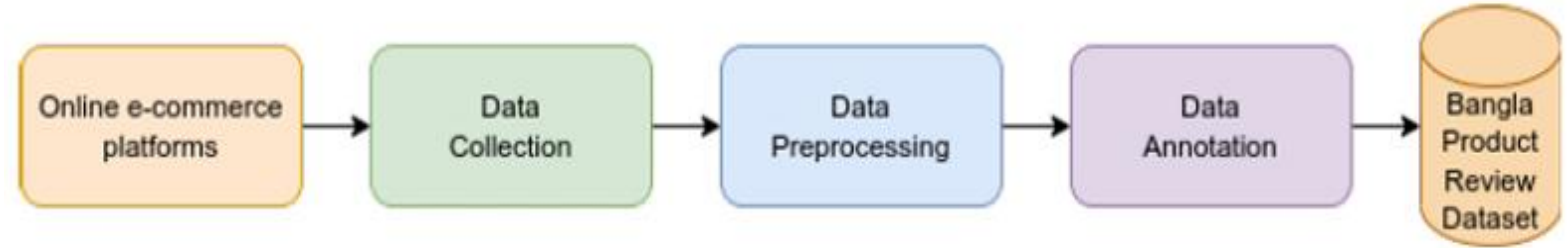


Figure 1: Data collection pipeline from online platforms to final dataset.

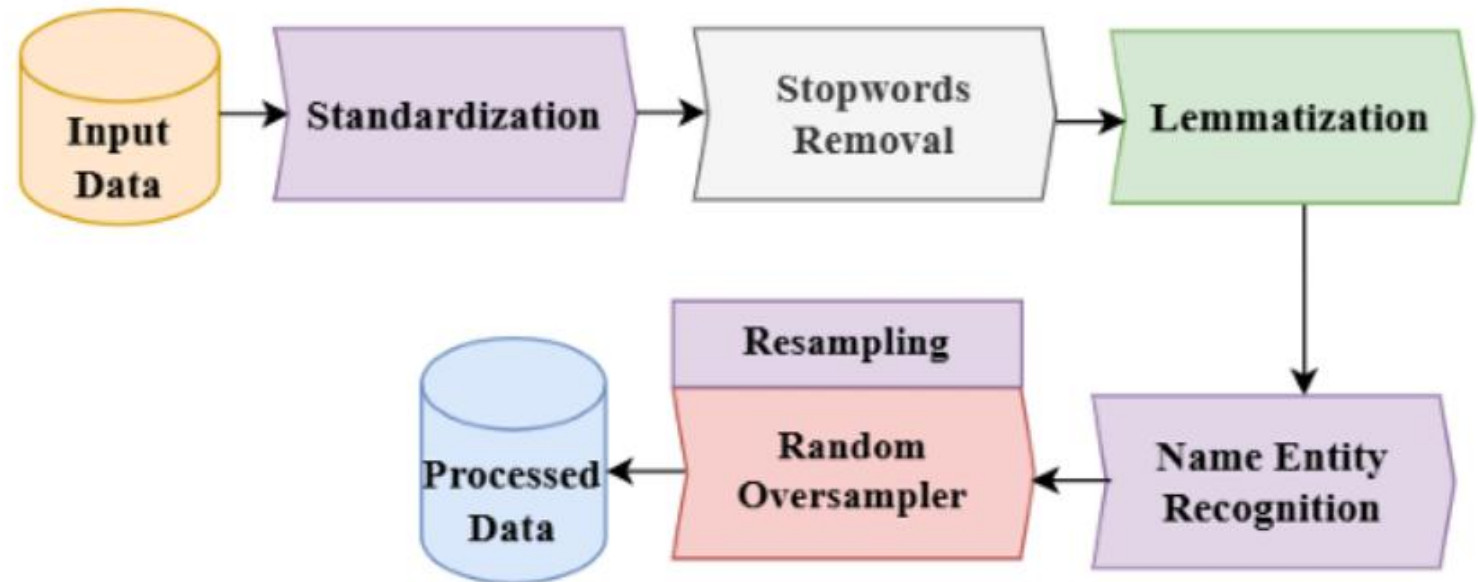


Figure 2: Data preprocessing pipeline from raw input to processed dataset

Vocabulary Overlap

- Vocabulary is rich and varied across aspects.
- Quality reviews possess the largest lexicon (more unique words).
- Semantic overlap between aspects quantified by Jaccard similarity.

For example, Pricing and Quality aspects share a 0.52 Jaccard similarity.

Table 3: Jaccard Similarity Coefficients (Aspects)

	Quality	Service	Pricing	Decoration
Quality	1.00	0.38	0.52	0.45
Service	-	1.00	0.41	0.58
Pricing	-	-	1.00	0.50
Decoration	-	-	-	1.00



- **BanglaBERT**: 12-layer transformer (768-dim)
- **BiLSTM**: 2 layers, 128x2 units (bidirectional context)
- **LSTM**: 2 layers, 256 units (sequential dependencies)
- **GRU**: 2 layers, 200 units (efficient gating)



Embedding Layer



Static Word Embeddings

GloVe (300-d) trained on 2.5B Bangla tokens for semantic content.



Dual Embeddings

Combines static word embeddings with FastText for comprehensive semantic capture.



FastText Embeddings

Captures subword information for morphological details.



BanglaBERT Contextual Embeddings

capture dynamic meanings of words based on sentence context.

Training and Ensemble

- **Individual Training:** BanglaBERT with AdamW ($\text{lr}=2\text{e-}5$), RNNs with Adam ($\text{lr}=1\text{e-}3$).
- **Adaptive Weighted Voting:** Weights learned via gradient descent on validation data.
- **Dynamic Weight Tuning:** Weights adapt based on review length and complexity.
- **Final Decision:** Softmax for multiclass, calibrated threshold (0.5) for binary.

Adaptive Learning



Experimental Setup

Dataset Split:

- Training: 70% (6,128 reviews)
- Validation: 15% (1,313 reviews)
- Test: 15% (1,314 reviews)

Evaluation Metrics:

- Accuracy, Precision, Recall, F1-Score
- Macro-average for balanced multi-label evaluation

Implementation:

- Python 3.8, TensorFlow 2.4
- Individual model training followed by ensemble optimization

Baseline Model Performance

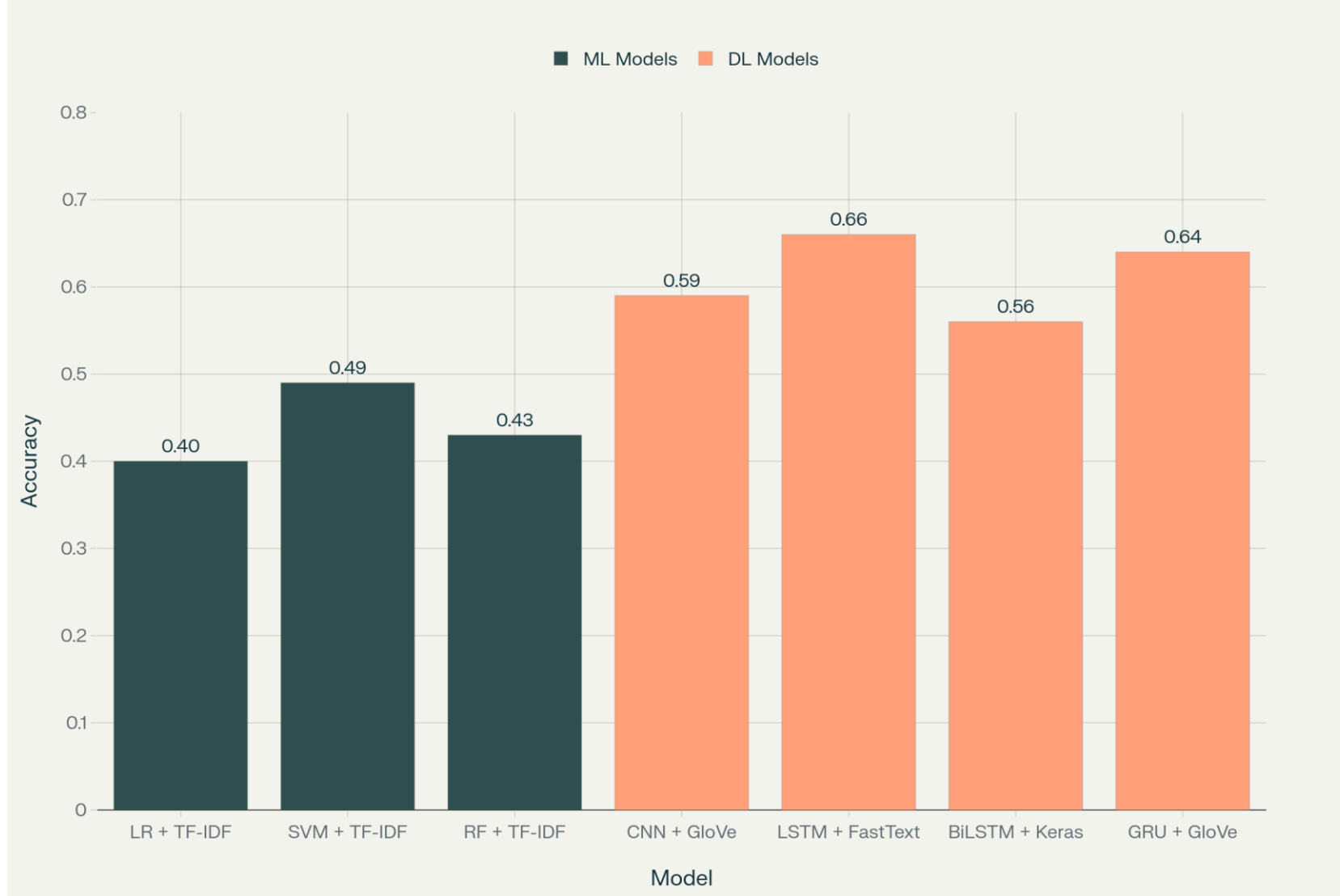


Figure 4: Performance Comparison of Baseline Models

BanglaSentNet Performance

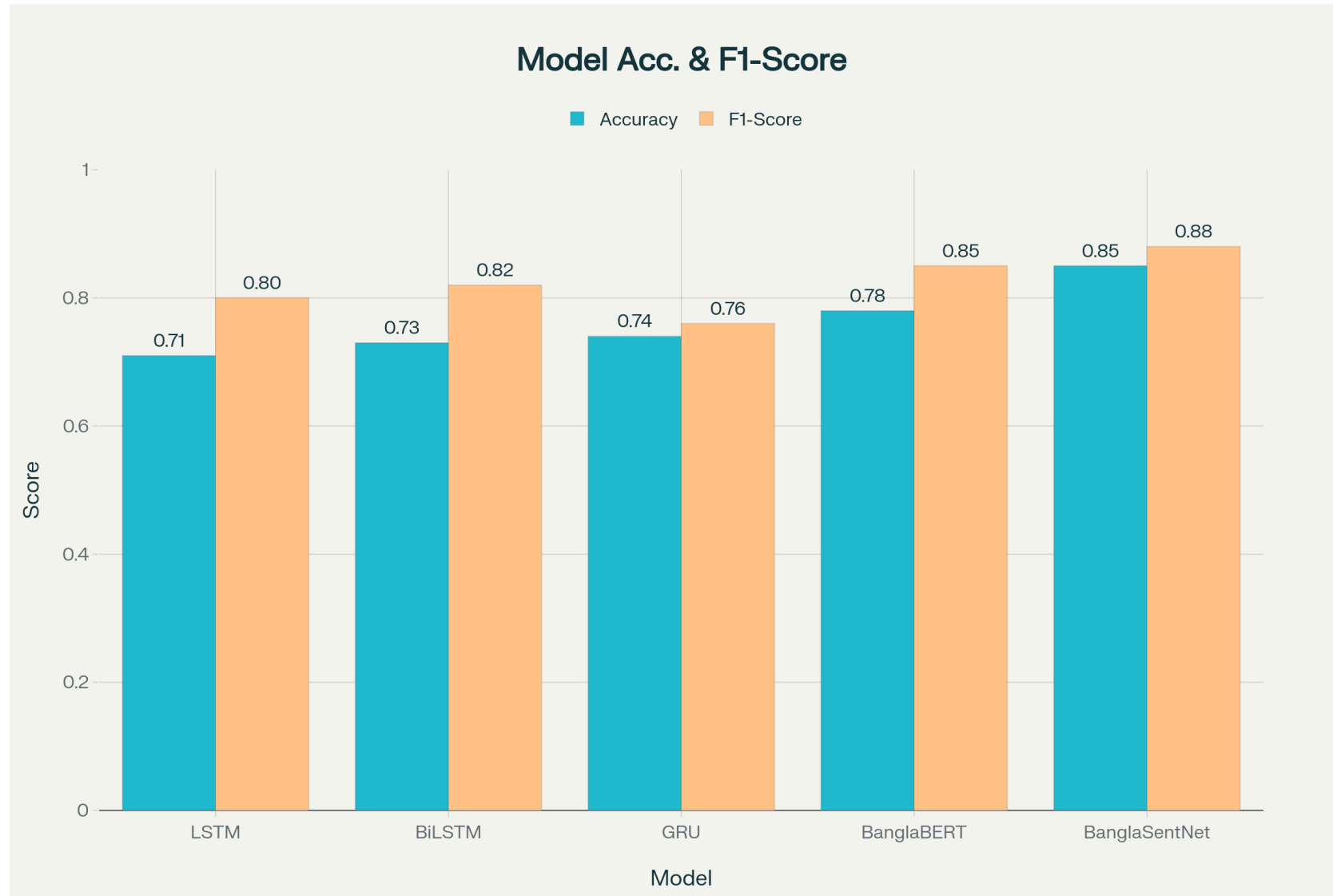


Figure 5: Performance Comparison of Final Models

Ablation Study

Full Ensemble

F1: 0.88

Without BanglaBERT

F1 drops to 0.75

Without BiLSTM

F1 drops to 0.79

Without LSTM

F1 drops to 0.81

Without GRU

F1 drops to 0.78

Removing any component degrades performance, highlighting the value of each model.

Error Analysis

- **Major Error Types:** Complex negation, sarcasm, implicit sentiment, spelling variations, context-dependent polarity.
- **Most Frequent Error:** Context-dependent polarity (same term, different sentiment).

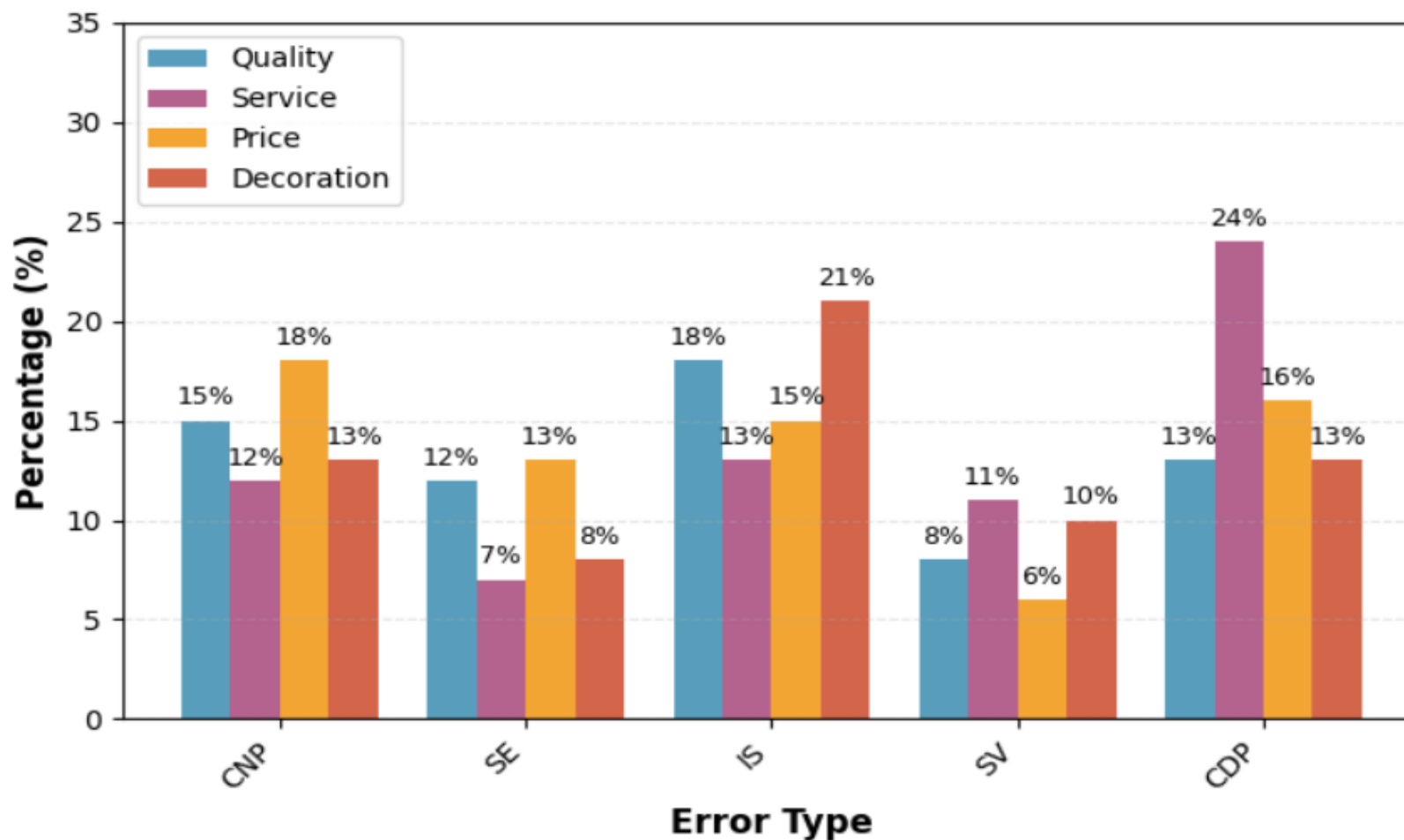


Figure 6: Distribution of Error Types Across Product Categories

Confusion Matrix

- Most positives (600) and negatives (450) correctly detected.
- Neutral sentiment is the hardest (263 correct), often confused as positive or negative.
- Typical in sentiment tasks due to implicit tones.

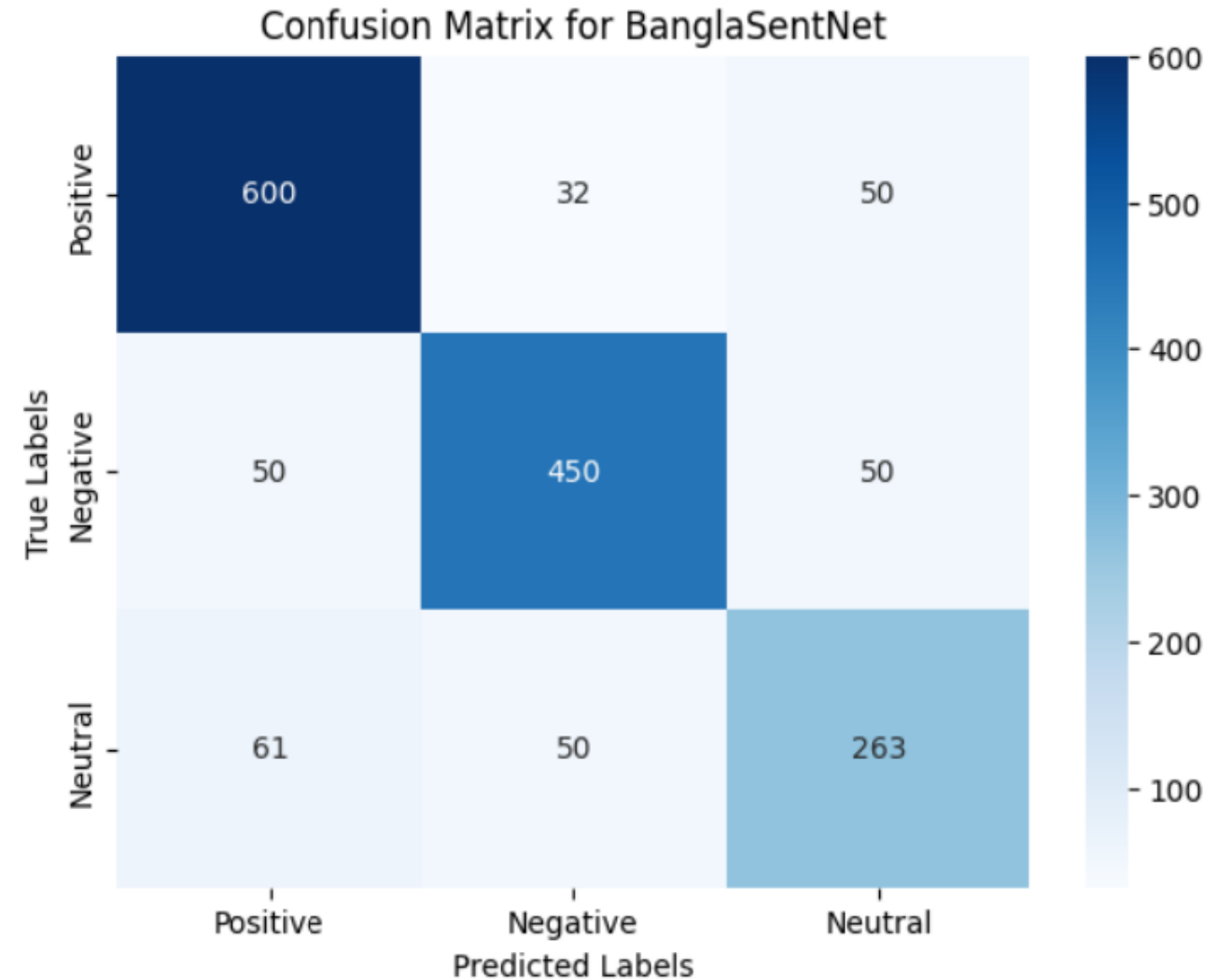


Figure 7: Confusion Matrix for BanglaSentNet on Test Data

Real-World Case Study



E-commerce Data

Applied BanglaSentNet to live product feedback (Daraz).



Actionable Insights

Detected high positive sentiment on quality, but negative on pricing, leading to strategy change.



Practical Value

Improves consumer insight and decision-making.

Conclusion and Future Work

Conclusion:

- **BanglaSentNet** improves multi-aspect sentiment analysis for Bangla reviews.
- Tested on a new dataset of 8,755 Bangla reviews.
- Achieved **85% accuracy** and **0.88 F1** on an 8,755-review dataset, outperforming baselines.
- Sets new state-of-the-art for low-resource Bangla SA.

Future Work:

- Optimize ensemble to reduce compute cost; explore few-shot/transfer learning.
- Address limitations (informal text, slang) and extend to more domains.

References

- [1] Pontiki et al. (2015), SemEval-2015 Task 12: Aspect based sentiment analysis
- [2] Pontiki et al. (2016), SemEval-2016 Task 5: Aspect based sentiment analysis
- [3] Cai et al. (2020), Aspect-category based sentiment analysis with hierarchical graph convolutional network
- [4] Maroof et al. (2024), IEEE Access: Aspect-based sentiment analysis for service industry
- [5] Mandal et al. (2014), Supervised learning methods for bangla web document categorization
- [6] Dhar et al. (2018), Performance of classifiers in bangla text categorization
- [7] Hasan et al. (2017), Multi-label sentence classification using bengali word embedding model
- [8] Hoq et al. (2021), Sentiment analysis of bangla language using deep learning approaches
- [9] Islam et al. (2020), Sentiment analysis in bengali via transfer learning using multi-lingual bert
- [10] Kabir et al. (2023), BanglaBook: A large-scale bangla dataset for sentiment analysis
- [11] Hoque et al. (2024), Exploring transformer models in sentiment analysis for Bengali language
- [12] Chakma et al. (2023), Low-resource sentiment analysis leveraging BanglaBERT
- [13] Mahmud et al. (2024), Enhancing sentiment analysis in bengali texts: A hybrid approach

Thank You!

Questions?

Presenter: Md Rifat Hossen

Chittagong University of Engineering and Technology

Email: rifat8851@gmail.com

